Lab 1 Use-Case Flow

Analyze Slow Queries and Recommend Indexes | Problem Statement 44

Primary actor: Database Administrator. Supporting actor: Slow-query log source.

Preconditions: At least one configured PostgreSQL or MySQL source is available; slow-query logging and plan collection are enabled; the user is an authenticated Database Administrator.

Postconditions: Each valid query is linked to its parsed plan; supported missing-index candidates and evidence are saved for review; no database index is changed automatically.

Main success scenario

1. The DBA opens the profiler and selects a source and analysis window.
2. The profiler ingests slow-query records and validates required fields.
3. The profiler parses each available EXPLAIN plan and associates it with its normalized query.
4. The profiler identifies sequential scans on large tables and analyzes filter and join columns.
5. The profiler creates and ranks non-duplicate index candidates, showing the supporting plan evidence.
6. The DBA reviews a candidate and approves, rejects, or defers it; the profiler records the decision and timestamp.

Alternate flow - plan cannot be parsed

3.a At step 3, a record contains an unsupported or malformed execution plan.

3.b The profiler records the query identifier and parse error, excludes that plan from index recommendations, and continues processing other valid records.

3.c The results page reports the skipped record count and offers the error details for correction.